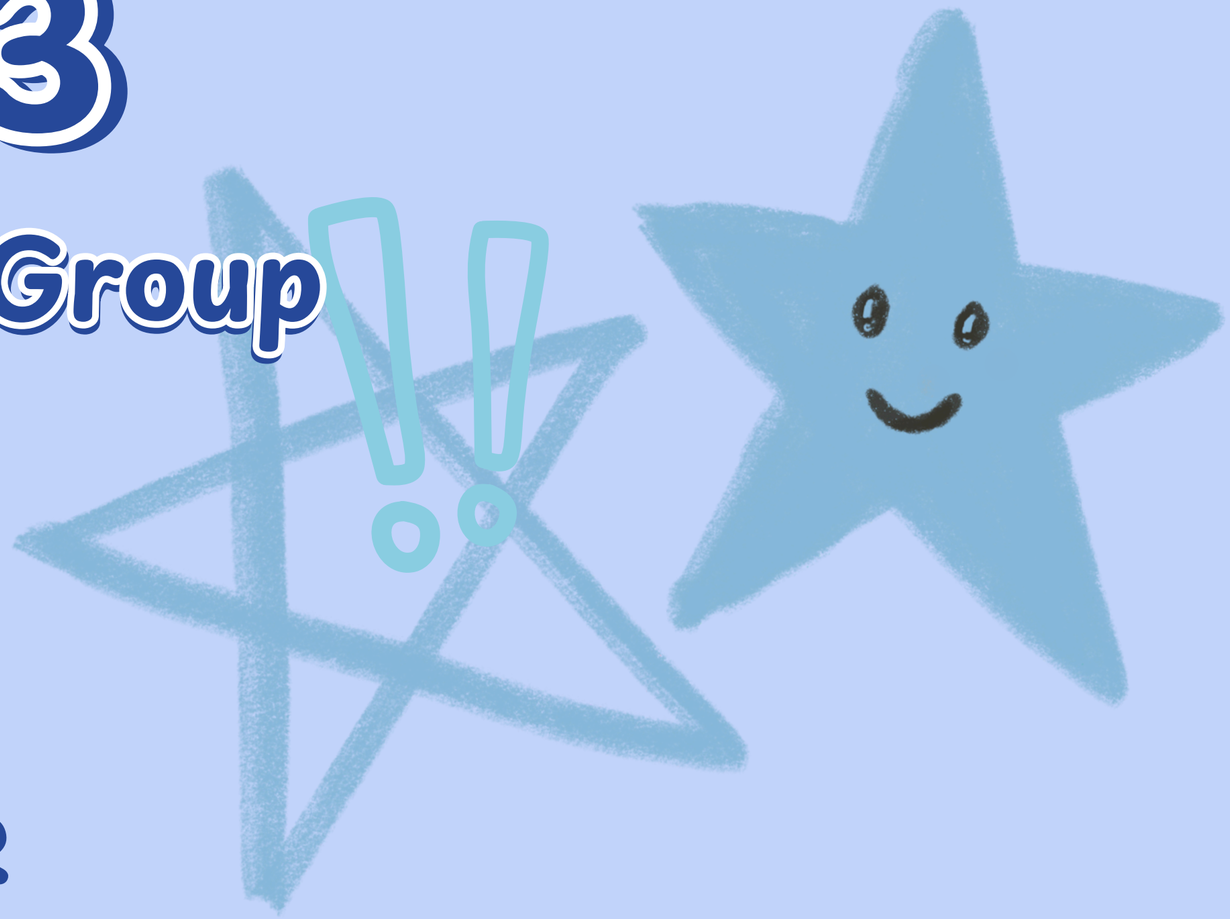
GROUP 3

Members of the Group

- Arkana
- Agil
- Putra
- Wafiq
- Fatih
- Ziad
- Zhahir
- Rizky R



“Theoritical Observasion On Waste Management”

A. Theoretical Framework

Waste is discarded materials from human activities that are no longer used. According to Law No. 18 of 2008 regarding waste management, waste management aims to improve public health and environmental quality.

Theoretically, waste is divided into 3 types:

* Organic: easily decomposable (food scraps, leaves, and vegetables) — 60% of waste in Indonesia
* Inorganic: difficult to decompose (plastic, glass, metal, and cloth)
* Hazardous and Toxic Waste: hazardous (batteries, medical waste, electronic waste)

B. Commonly Used Processing Theory

1. The 3R+R Concept

- Reduce: reduce waste at the source
- Reuse: use again (bottles become pots)
- Recycle: reprocess into new products
- Replace: replace non-environmentally friendly materials with eco-friendly ones

2. Source Management (Household Level)

- Sorting: separate Organic and Inorganic waste using 2 bins
- Waste Bank: circular economy system; inorganic waste has value when collected
- Composting: organic waste decomposed into compost (Takakura Method, Biopori, Black Soldier Fly/BSF Maggot)

Waste Management

Waste management is a series of activities to control waste from its source up to the final disposal site.

October 12, 2026

Types of Waste

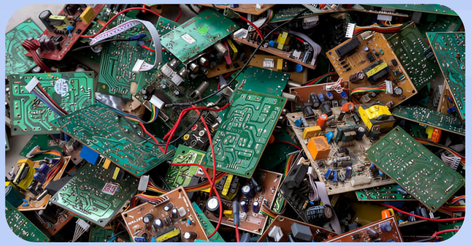
Organic, inorganic, and electronic waste are types of waste categorized by their origin and nature. Organic waste originates from living organisms and decomposes easily; inorganic waste is difficult to decompose naturally; and electronic waste comes from electronic devices that are broken or no longer in use.



Inorganic Waste:
food scraps, leaves.



Inorganic Waste:
plastic, metal.



Electronic Waste:
Capacitor, Resistor.

Observation of Waste: Organic, Non-organic, and Electronic

Observations of the surrounding environment show that waste falls into three categories: 1. Organic Waste — originates from living organisms and decomposes easily. Examples: food scraps, fruit peels, and dried leaves. It can be processed into compost.

2. Non-Organic Waste — does not decompose easily but can be recycled. Examples: plastic, paper, cans, and glass. Ideally, this should be sorted and taken to a waste bank.

3. Electronic Waste — discarded electronic items containing hazardous materials. Examples: broken mobile phones, batteries, cables, and adapters. It must not be disposed of carelessly; it should be handed over to specialized waste management facilities. Conclusion: Sorting waste at home is crucial to ensure proper handling and to keep the environment clean.

Prinsip 3R (Reduce, Reuse, Recycle)

The 3R concept is an effective way to reduce waste accumulation. By reducing, reusing, and recycling, we can extend the lifespan of items and alleviate the burden on landfills.

3

Reduce:

reduce plastic use

Reuse:

reuse containers

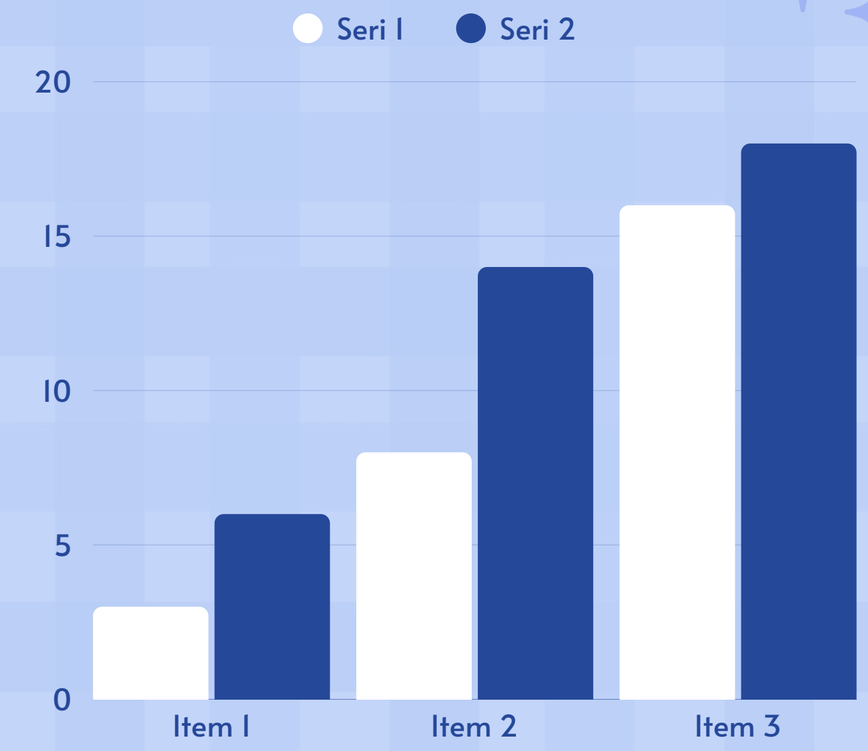
Recycle:

recycling used goods

Waste Management in Schools

Schools can serve as important venues for education on waste management. Through activities such as waste banks, recycling competitions, and cleanliness programs, students are encouraged to care for the environment from an early age

Innovation in Waste Management



- Many innovations have been developed, such as ecobricks, biodigesters, and digital applications for waste collection services. These innovations make waste management more efficient and environmentally friendly.

The Role of Government and Policy

- The government plays a crucial role in formulating regulations, providing infrastructure, and raising public awareness through sustainable waste management campaigns.

Positive Impacts of Waste Management

When waste is managed effectively, the impact is immense: the environment becomes cleaner, the creative economy grows, and communities lead healthier lives. Waste can be a source of economic value.



Let's Manage Waste Wisely!

01. Start from home

02. Be an agent of change.

03. Start from home

Thank You